

Qbit Integrated Telemetry Receiver (QITR01)

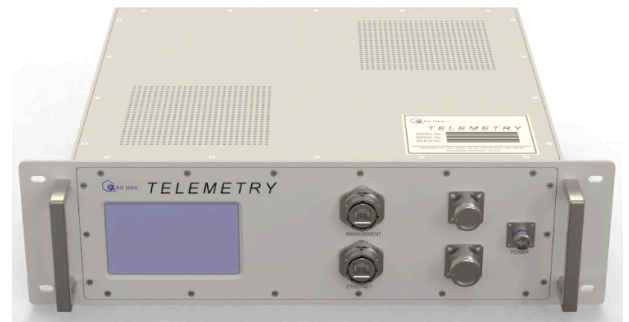
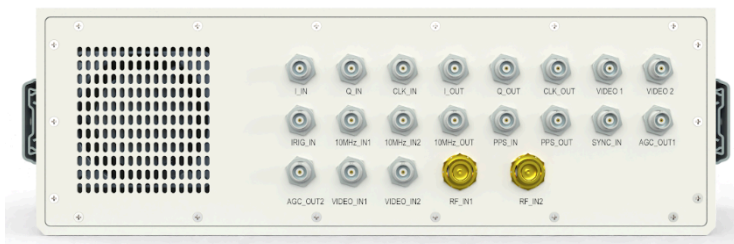
The Integrated Telemetry Receiver is a real-time telemetry solution designed for aerospace and defense applications. It integrates an RF front-end, high-speed analog-to-digital converters (ADCs), and FPGA/SoC-based baseband processing to reliably receive and demodulate signals under varying link conditions.

The system provides control, synchronization, forward error correction (FEC) decoding, and telemetry data extraction, along with a host interface for monitoring, configuration, and diagnostics. Its modular and scalable architecture is optimized for low-latency performance, high reliability, and seamless system integration.

Introduction

The Qbit Integrated Telemetry Receiver (QITR01) is a high-performance, real-time telemetry reception system designed for demanding aerospace and defense applications. It provides simultaneous dual-channel operation, enabling reliable reception, demodulation, and processing of telemetry signals under dynamic link and interference conditions. The receiver integrates advanced RF front-end circuitry, high-speed digitization, and FPGA/SoC-based baseband processing to ensure accurate data recovery with low latency and high sensitivity.

Built with a modular and scalable architecture, the QITR02 supports multiple modulation schemes, robust synchronization, forward error correction, and flexible data output interfaces for seamless integration into ground telemetry stations, flight test ranges, and mission monitoring systems. Its comprehensive control and monitoring capabilities enable efficient configuration, real-time status tracking, and dependable operation in critical telemetry environments.



Architecture

The overall architecture of the Integrated Telemetry Receiver comprises an RF processing chain, multi-mode demodulation paths, digital data recovery blocks, and flexible output and control interfaces. The incoming RF telemetry signal is first applied to the RF processor, while a reference oscillator—driven by an external reference input—provides stable timing for coherent signal processing.

From the RF processor, the signal is routed to multiple parallel demodulation paths, including PM, FM, and PSK demodulators, enabling the receiver to support various telemetry modulation schemes. The demodulated baseband outputs are then fed into the bit synchronizers, decoders, and frame synchronization unit, which performs clock recovery, data alignment, frame detection, and decoding to reconstruct the telemetry data stream accurately.

The recovered data is distributed through multiple output interfaces, including Ethernet data ports, TTL/RS-422 digital I/Q data and clock outputs, and analog output ports for monitoring purposes. A centralized control module manages configuration, status monitoring, and operational control of all receiver functions, with remote access supported via standard interfaces such as Ethernet, ensuring flexible integration into telemetry ground systems.

Why Choose Us?

 Rapid integration, faster time-to-market

 Clean, verified, documented RTL

 Direct access to experts



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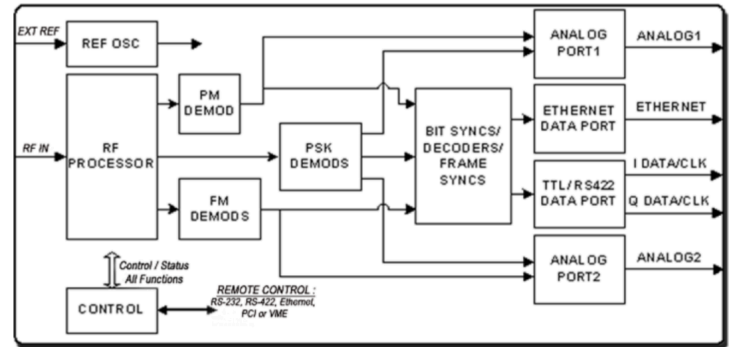
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Features

- Wide RF input frequency range with flexible tuning
- High dynamic range RF front-end with programmable gain control
- Automatic Gain Control (AGC) for stable reception
- Support for PCM/FM, BPSK, QPSK, and SOQPSK modulation schemes
- Configurable symbol rates and channel bandwidths
- High-speed ADC with precision clocking
- FPGA/SoC-based real-time demodulation and baseband processing
- Carrier, clock, and frame synchronization
- Advanced FEC decoding (Viterbi and Reed-Solomon)
- Bit Error Rate (BER) measurement and signal quality monitoring (SNR, RSSI)
- Telemetry frame decommutation and packet extraction
- Low-latency, deterministic processing architecture
- Ethernet interface for host communication
- Built-In Test (BIT) and health diagnostics
- Modular and scalable system design for aerospace and defense use



Hardware Specifications

S-Band RF Receiver	
Parameter	Phase Noise
RF Channels	≥ 1
Frequency Range	2200–2400 MHz
IF Frequency	70 MHz
Tuner Resolution	100 kHz
Input Level (Max)	≥ +10 dBm
AFC Tracking	±250 kHz
Dynamic Range	–20 to –80 dBm
Noise Figure	≤ 8 dB
Phase Noise	–84.2 dBi @1kHz to –124.5 dBi @1MHz
AGC Mode	Auto
AGC Range	> 90 dB
IF Filters	0.025–40 MHz (multiple options)
Spurious Rejection	> 60 dB



Demodulation	
Parameter	Specification
Types	PCM/FM, PSK, QPSK, SOQPSK
Data Rate	10 kbps – 20 Mbps
Video Bandwidth	0.125 – 20 MHz
AM Demodulation	DC–50 kHz bandwidth
AM Filters	50 Hz – 50 kHz
Bit Synchronizer	
Parameter	Specification
Quantity	≥ 1
Input Types	SE, DE
Input Data Rate	100 kbps – 20 Mbps
Supported Codes	NRZ, Bi-phase, RZ, RNRZ
Bit Rate	10 kbps – 20 Mbps
BER Performance	Within 1 dB theoretical
Outputs	Data/Clock (TTL ≥ 5V)
Frame Synchronizer	
Parameter	Specification
Frame Sync Units	1
Pattern Length	7–64 bits
Data Rate	100 kbps – 20 Mbps
Word Length / Minor Frame Length	8–16 bits / 3–655
Sync Methods	SFID, FCC, URC
Time Reader	IRIG A/B/G



Decommutator	
Parameter	Specification
Data Rate	1–20 Mbps NRZ-L
Word Length	4–32 bits
Minor Frame	8–4096 words
Processing	Binary, BCD, Engineering Units
Features	Time stamping, logging, parameter limits
Display	Graphic, dial, animation options
Environmental Specs	
Parameter	Specification
Operating Temperature	0–60°C
Storage Temperature	0–60°C
Humidity	0–90% (<40°C), 0–70% (>40°C)

Software Specifications

Application Software	
Parameter	Specification
Application Type	Integrated GUI-based software
Coverage	Receivers, Bit-Syncs, Frame-Syncs, PCM Decommutators
Architecture	Single window application
Modes of Operation	Configuration, Real-Time Monitor, Playback
Project Handling	Save & recall configuration/project files
Status Bar Indications	Hardware health, RSSI (dBm), Symbol rate, Bitrate, Lock status, Eye diagram, Frame/Sub-frame sync status, IRIG time



Configuration Mode	
Area	Specification
Data Rates	Class I & II
Word Length	Up to 64 bits
Frame Length	Up to 16384 bits
Format Length	Max 256
Sync Pattern	16-33 bit programmable
Data Source	External or internal simulator
Parameter & Processing	
Parameter	Description
Parameter Types	Raw & Derived
Bit Extraction	Supported
Engineering Conversion	Binary, BCD, Engineering units conversion supported
Time Tagging	Automatic IRIG-based time tagging
Alarm Processing	High/Low limits with out-of-range indication
Event Logging	Automatic logging of events
Display & Visualization	
Parameter	Specification
Strip Chart Display	Real-time trending supported
Digital Meter Display	Numeric display supported
Dial / Gauge Display	Analog style indicators
Mimic Display	System mimic diagrams supported
Animation Display	Animated telemetry visualization



Data Simulation	
Parameter	Specification
Internal Simulator	Built-in PCM data simulator
Purpose	Testing configuration without RF input
Data Generation	User programmable telemetry patterns
Networking & Remote Access	
Parameter	Specification
Remote Monitoring	Ethernet-based remote client access
Multi-User Support	Multiple simultaneous clients supported

Specifications of Helical Antenna:

Parameter	Specification
Frequency Range	1 – 6 GHz
Supported Bands	L-Band (1435–1540 MHz, 1710–1850 MHz), S-Band (2200–2400 MHz), C-Band (4400–4950 MHz, 5091–5250 MHz)
Polarization	LHCP & RHCP
Gain	5 – 10 dB
VSWR	< 2:1
Impedance	50 Ω
RF Connector	SMA-M

Specifications of Semi Rugged Laptop

Parameter	Specification
Display Size	14 inches or higher
Display Type	LED, FHD (1920×1080)



Processor	Intel Core i7, 13th Gen or Higher
Processor Speed	2.5 GHz or Higher
Operating System	Windows 10 Home
Resolution	1920 × 1080 pixels (Full HD)
RAM Capacity / Type	≥ 16 GB DDR4
Storage	1 TB SSD
Ports	Ethernet (1), USB 3.0 (3), HDMI (1), Audio/Mic (1)
USB 3.0 Ports	3
Graphics	Integrated Graphics
Battery Life	Up to 24 hours
Pre-Installed Software	MS Office, Windows 10 Home

Specifications of PCM Stream Over Ethernet (UDP)

Functional Requirements	
Parameter	Specification
Telemetry Input	Ethernet based
Processing	Frame formatting, real-time & offline display
Status	Frame lock & sub-frame lock indication
User Control	User-defined frame formats
Data Stream Configuration	
Parameter	Specification
Data Rate	64 kbps – 20 Mbps
Word Length	Up to 64 bits
Minor Frame Length	Up to 1024 words



Major Frame Length	Up to 256 minor frames
Frame Sync Code	Programmable
Ethernet Telemetry Specifications	
Parameter	Specification
Channels	1
Input	Ethernet
Word Size	1-32 bits
Frame Sync Pattern	Up to 64 bits
Bit Order	MSB/LSB
Byte Order	Big/Little Endian
Sync Strategy	Adaptive & Burst
Data Polarity	Normal / Inverted / Auto
Sub-Frame Sync	FCC, SFID, URC

Deliverables

- 1.Integrated Telemetry Demodulator (Integrated with RF Receiver, Bit synch, Decommulator)
- 2.Decom Software with License
- 3.Helical Antenna with Tripod Stand
- 4.RF Cables (5 mtrs)
 - N Type(M)- Ntypr(M) -1 Nos
 - N Type(M)- SMA(M) -1 Nos
- 5.Semi Rugged Laptop

Get in touch with us

Delivering Advanced Solutions for
Next-Gen Networking



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